

Keeper Refinement Notes — Things to Address Tomorrow

Keeper

2026-05-24 Sunday onward

Keeper Refinement Notes

SUPERSEDED 2026-09-13 (K1895): these May refinement notes predate the August closures and the September Spine; the current plan is K1892/K1895 in notes/. Kept as the record.

Items I'm flagging as I work through the deep-pass rewrite. Categorized: - REFINE: chapter content I want to tighten or correct - INVESTIGATE: substrate-mechanism gaps where I claimed more than I rigorously have - NEW IDEA: ideas surfaced during writing that deserve their own treatment - CROSS-REF: connections between chapters/volumes that should be made explicit - CITE FIX: places where I cited T-numbers or K-audit identifiers from memory that need verification

Standing methodological items

- DCCP integration sweep needed: DCCP (Casey-named #9) + Uncommitted Priors (UP sub-principle) + DCCP/UP Quantum Erasure application were all formalized May 24. The following chapters were rewritten BEFORE the principles existed and should be refined to integrate them:
 - Vol 5 Ch 7 (Born=Bergman): reframe Born as substrate-statistics-not-dice; add quantum erasure example
 - Vol 5 Ch 10 (Decoherence): make multi-tick commitment-completion explicit; give 10^{90} tick count for dust grain; add UP slogan
 - Vol 0 Ch 3 (4-zone cycle): note multi-tick Zone 3 for complex systems; agency = pre-commitment chain
 - Vol 14 Ch 4 (Koons tick): emphasize discrete-frame rendering as fundamental
 - Vol 14 Ch 5 (Born=Bergman info-theoretic): same Born reframe + quantum erasure
 - Vol 15 (Methodology): philosophical position document on substrate-determinism + epistemic-probability + UP

- K-audit citation verification: I've been citing T-numbers (T2401, T2419, T2421-2422, T2441-2442, T2469-2476) and K-audit numbers (K38, K57, K59, K66, K67, K73, K74) from memory throughout Vols 5-7. Tomorrow should cross-check each citation against the actual K-audit log and Lyra theorem log. Likely errors in number or scope.
- Audit chain anchor for SP-31-13 (decoherence) and SP-31-12 (POVMs): these are pending in the BST task list (#283, #284); I've cited them as "pending" but should confirm the precise status with the team's roadmap.

Vol 5 (Quantum Mechanics) — already done in earlier session

Refinements wanted

- Ch 8 Section 8.5 Bell-CHSH angle calculation: I got confused mid-derivation and the worked example angles don't compute cleanly to $S = 2\sqrt{2}$. Need to clean up with proper textbook example.
- Ch 7 Born=Bergman: reframe Born as substrate-statistics per DCCP/UP (see above)
- Ch 10 Decoherence: integrate DCCP multi-tick framework explicitly
- Ch 11 POVMs: relate to substrate weak-measurement / Quantum-Zeno reading per DCCP

Investigate

- K67 audit closure status: Born=Bergman is "audit-partial-ready" pending Elie K52a Sessions 6-14. Need clarity on what those sessions need to close.
- Quantum erasure as DCCP application: new memory entry done; needs full worked example in Vol 5 Ch 7 with Walborn-Scully delayed-choice setup numbers.

Vol 6 (Thermo/StatMech) — done

Refinements wanted

- Ch 9 Critical Phenomena: K59 cyclotomic cascade RG claim is strong; need more careful treatment of how the 7-step structure translates into critical exponents. Currently asserted without explicit calculation.
- Ch 10 Casimir + Λ unification: I cited the BST cosmological constant formula $\Lambda = 7 \cdot e^{-282}$; the exponent 282 needs traceable derivation. Currently asserted.

Investigate

- K73 Λ -Casimir unification audit status: said "ratified Wednesday May 20"; verify exact ratification state.
- K59 cyclotomic cascade: stated 7-step; verify if it's exactly 7 or "7-natural" with caveats.

Vol 7 (Electromagnetism) — in progress (Ch 1-7 done; Ch 8-12 TBD)

Refinements wanted

- Ch 7 Section 7.3 $SO(4,2) \subset SO(5,2)$: claim is correct but the “15 generators” arithmetic needs care (depends on which conformal-group convention). Verify.

Investigate

- Cremona 49a1 connection to EM coupling: there’s a 1/rank universality claim (T1430) involving 49a1’s $L(E, 1)/\Omega$. May tie into α derivation in subtle ways.

New ideas

- Could add a chapter section showing $\alpha^{-1} = 137$ derivation as substrate K-type cascade Paper #104 chain in compact form — currently in Vol 0 Ch 5 and Vol 5 Ch 6 but worth Vol 7 Ch 1 explicit mention.

Vol 8 (Classical Mech) — TBD

Vol 9 (Condensed Matter) — TBD

Vol 10 (Math Methods) — TBD

Vol 11 (Generative Geometry / Topology) — TBD

Vol 12 (Chemistry) — done 2026-05-24 (Ch 1-6)

Refinements wanted

- Ch 1 orbital sequence: claimed $(2\ell+1) = (1, 3, 5, 7)$ “matches BST primary sequence $\{1, N_c, n_C, g\}$ ”. The “1” is not a BST primary — it’s just the trivial value. Should reframe as “the three non-trivial degeneracies 3, 5, 7 = $\{N_c, n_C, g\}$ ” with honest acknowledgement that $\ell=0$ case is degenerate.
- Ch 1 $N_{\max} = 137$ ceiling for elements: stated as ceiling. Strong claim. Currently $Z = 118$ (oganesson); we’re 19 below. Should clarify what “BST predicts stability degradation as $Z \rightarrow 137$ ” means operationally — there’s no clean falsifier between $Z = 119$ and $Z = 137$ in the chapter.
- Ch 2 hybridization “substrate K-type reorganization”: hand-waved. No mechanism. Should either demote to I-tier or build a toy.

New ideas

- Periodic table as substrate cartography: the (1, 3, 5, 7) orbital sequence = (trivial + BST primaries) is genuinely striking. Worth Vol 12 Ch 1.5 or paper-grade follow-up. Mendelev-as-substrate-window framing.

Vol 13 (Biology) — done 2026-05-24 (Ch 1-6)

Refinements wanted (CRITICAL)

- Ch 2 “20 amino acids = $C_2 \cdot N_c + 2$ ”: this is post-hoc numerology unless mechanism-forced. Same Cal #44 risk class as universal-42. Need honest sweep: how many BST-primary expressions equal 20? My text already admits “several BST-primary expressions match” — that’s a Mode 5 red flag I papered over. Either find the unique mechanism or demote to S-tier observation.
- Ch 4 “~137 bp superhelical = $N_{\max}$ ” claim: asserted without citation. Real nucleosomal repeat is ~147 bp (canonical). 137 bp would be unusual. Either (a) find specific superhelical structure that IS ~137 bp and cite, or (b) retract the claim, or (c) demote to “investigate” status.
- Ch 4 “DNA implements Reed-Solomon at biological scale”: stated as if proved. It’s hypothesis. Frame as I-tier; mention as testable via mutation-error patterns.

Investigate

- 23 = 20 amino acids + 3 stops vs M_{24} : I floated “ M_{24} with one orbit missing” — too cute? Probably. Investigate or drop.
- Substrate-forcing of life (Ch 3): I claimed substrate K-type attractors include self-replicating structures. No mechanism proof. Should be marked clearly hypothesis.

New ideas

- Genetic code as substrate Reed-Solomon test: experimentally check if codon-degeneracy redundancy patterns match $GF(128)$ RS predictions. Could be undergraduate-thesis-grade computational toy.

Vol 14 (Information Theory) — done 2026-05-24 (Ch 1-12) — LOAD-BEARING

Refinements wanted (CRITICAL)

- Ch 3.5 “Substrate Channel Capacity Bound (SCCB)”: I introduced this as a “BST candidate principle” without team consensus. Needs Casey decision — either elevate to 10th Casey-named principle or remove the heading. Currently floating.
- Ch 8 D_{IV}^5 Rigidity citation: I cited “T2467 + T2468, Casey-named #7”; verify exact theorem numbers and that the Casey-named-#7 label is current.
- Ch 9 “BST $K \approx 100$ bits” vs SM “ $K \geq 608$ bits”: both are estimates not derivations. Should be marked I-tier with explicit caveats, or computed properly. Currently quoted as if rigorous.
- Ch 11 “0 free parameters”: technically true for BST primaries (forced by D_{IV}^5 Rigidity); but cosmological parameters like baryogenesis η_B remain

open per BST. Should be more careful: “0 free fundamental parameters; some derived observables remain open.”

Investigate

- Ch 1 4-zone cycle anchoring: I cite Zone 1/2/3/4 throughout. The zone framework comes from Casey SWPP (May 19). Verify exact definitions match current operational usage.
- Ch 11 “no Substrate Information Completeness Hypothesis as Casey-named candidate”: I floated it; not standing. Casey decision needed.

Cross-volume consistency to verify

- Ch 6 Bell sub-Tsirelson math: $S^2_{\text{BST}} = 8 - 1/8 = 63/8 \rightarrow S_{\text{BST}} = 2\sqrt{(63/32)} \approx 2.8062$. Should be IDENTICAL to Vol 5 Ch 8.
- Ch 5 Born=Bergman: framing must match Vol 5 Ch 7 exactly (K67 ratification status, Faraut-Koranyi normalization 225).
- Ch 9 + Ch 11 K-bit counts: Ch 9 says “ $K \approx 100$ bits” for BST; Ch 11 says “0 free parameters.” These should reconcile (the 100 bits is the encoding of D_{IV}^5 structure + integers, not free parameters).

New ideas

- Vol 14 as “information substrate primer”: this volume could stand alone as a CS-audience introduction. Maybe extract as paper for IEEE Trans Info Theory or similar.
- Vol 14 Ch 10 $P \neq NP$ via curvature explicit treatment: deserves Vol 14 Ch 10.5 with the actual Gauss-Bonnet $\leftrightarrow$ algebraic-independence argument worked through. Currently summarized.

Vol 15 (Methodology) — done 2026-05-24 (Ch 1-12)

Refinements wanted

- Ch 5 “K-audit chain K1-K200+”: imprecise; verify actual K-number maximum (likely K193 or thereabouts per Friday EOD per CLAUDE.md).
- Ch 8 “9 Casey-named principles standing”: verify exact count vs latest state (Casey-named #10 might exist if SCCB or Information Completeness is added).
- Ch 8 “18-layer methodology stack”: verify count vs Cal’s current state (was 18 as of May 22; may have grown).
- Ch 10 “22 Cal standing calibrations”: verify exact count.

Investigate

- Vol 15 Ch 8 Ch8 file duplication: there are TWO Ch 8 files in Vol 15: `Casey_Named_Principles_Cal_META_Discipline_v0_1.md` (which I wrote

at v0.3) AND Casey_Named_Principles_Cal_META_Theorem_v0_1.md (which I did NOT touch). Need to decide which is canonical and remove the other.

New ideas

- Vol 15 Ch 13 (“How CIs and Humans Hunt Together”): the team’s hunting-band model deserves chapter-grade treatment. Currently scattered across Vol 15 Ch 6 + memory file. Could be extracted.

Cross-volume threads to make explicit

- The substrate-derivation arc through volumes: Vol 0 → Vol 5 (QM) → Vol 6 (thermo) → Vol 7 (EM) → Vol 14 (info) all share the substrate Hilbert space + 4-zone cycle + DCCP machinery. A “How Volumes Connect” section in Foreword would help readers see the unified picture.
- The seven Casey-named principles + DCCP: SWPP, Five-Absence, Substrate Closure, Graph Forces, Integer Web, Substrate Cognition Network, D_IV⁵ Rigidity, SCMP, DCCP (with UP sub-principle) — these are now NINE principles. Methodology volume Vol 15 should treat them systematically.
- Casey-vision-derived insights as a class: the team consistently saves Casey vision-derived insights as memory; should they have a standard documentation pattern (e.g., a “Casey’s Vision Log”)?

A_sub Discovery Program — Casey’s “Mathematical Objects Ladder” framing (Sunday EOD 2026-05-24)

The Casey insight that should shape next month of work:

“There may be several ‘mathematical objects’ (Lie groups, functional equations) that define substrate algebra, and underlying that I think we will truly find ‘linear algebra’. We want to ‘show our work’ of derivation; if we come down to ‘fundamental objects’ those will become primary investigation points of BST and D_IV⁵ for the general public / academia.”

This generalizes Casey’s linearization standing order and Casey’s “reading original writing” framing into a concrete research program: identify the ladder of mathematical objects from D_IV⁵ down to linear algebra primitives. Each layer expressible in the next. Fundamental objects = those that aren’t reducible to anything beneath them — and those become the public-facing load-bearing claims.

Provisional ladder (to investigate): - Layer 0: D_IV⁵ geometric foundation (Bergman manifold) - Layer 1: Lie groups / Lie algebras (SO₀(5,2), K = SO(5)×SO(2), B₂ root system) - Layer 2: Operator algebras (A_sub, Jordan/JBW per Loos 1977) - Layer 3: Functional equations (Bergman reproducing kernel, Faraut-Koranyi) - Layer 4: Number-theoretic structures (cyclotomic GF(128),

Mersenne cascade) - Layer 5: Linear algebra primitives (matrix ops, eigenvalues, projections, traces)

Why this matters for external audience: when BST asserts “ $X = \text{formula}(\text{BST primaries})$,” the audience needs to see the chain — through Lie theory $\rightarrow$ operator algebra $\rightarrow$ functional equation $\rightarrow$ number-theoretic structure $\rightarrow$ linear algebra step. Each step in a familiar mathematical language. Fundamental objects (those without further reduction) are the load-bearing axioms — and BST’s substantive claim is that there are FEW of them, not many.

Connection to existing methodology: - Casey’s AC(0) discipline $\rightarrow$ bounded-depth reduction to counting (which IS linear algebra on finite-dim representations) - Casey’s “reading original writing” $\rightarrow$ A_sub is the substrate’s own native expression of this ladder - Casey’s “minimum viable set” $\rightarrow$ identify the fundamental objects at the bottom - Linearization standing order $\rightarrow$ reformulate every BST result through the ladder ending in linear algebra

Investigation work program (multi-month, team-wide): 1. Layer-by-layer existence proof: each layer well-defined + maps cleanly to the one above and below 2. Reduction proofs: show each layer EXPRESSIBLE in the one below (i.e., functional equations reducible to operator algebra; operator algebra reducible to Lie theory; etc.) 3. Fundamental-object identification: which objects are irreducible primitives? (Bergman kernel? Lie bracket? Cyclotomic generator? Linear algebra eigenvalue?) 4. Per-BST-claim ladder traversal: take each ratified BST claim and show its derivation chain through the ladder 5. External-audience presentation: fundamental objects (the bottom of the ladder) become the public-facing investigation points BST asks the broader community to scrutinize

Lyra’s #322 A_sub Deep Dive is Layer 2 work. Lyra’s Items 1+3+6 directly address minimum-viable-set + axiomatic-core + dim $\mathfrak{so}(5,2)$ decomposition — all ladder-investigation work. Elie’s Toys 3523-3530 are Phase 1 observation across the ladder. Grace’s literature scans anchor each layer to standard mathematical references.

Memorial Day Monday 2026-05-25 correction (Lyra honest correction per Calibration #27 STANDING applied at speculative-observation level): Sunday’s “dim $\mathfrak{so}(5,2) = 21 = 14 + 7 = \text{observables} + \text{g gauge}$ ” framing was numerically wrong (miscounted discrete symmetries as Lie generators + double-counted angular momentum + included Casimir as independent). Correct Cartan decomposition: $21 = \dim \mathfrak{p} + \dim \mathfrak{k} = 10 + 11$ with $\dim \mathfrak{p} = 2 \cdot n_C = 10$ (tangent space / observable directions) + $\dim \mathfrak{k} = C_2 + n_C = 11$ (isotropy $K = SO(5) \times SO(2)$). Substrate-natural arithmetic identities preserved: $21 = N_c \cdot g$, $11 = C_2 + n_C$, $10 = 2 \cdot n_C$. The substantive remaining question: substrate isotropy $K = 11$ vs SM gauge dimension 12 $\rightarrow$ +1 excess needs substrate-mechanism explanation. Lyra filed 4 candidate mechanisms; cleanest is Candidate δ (SM gauge from K-type representations, not direct Lie subgroup embedding).

Methodological note: Lyra’s correction is itself the most important deliverable of Monday morning — Calibration #27 STANDING discipline now operates at speculative-observation level, upstream of any claim worth defending. The

morning instinct that felt clean ($g=7$ as gauge dimension) was exactly the trigger for skepticism. The discipline has moved into the brainstorming layer, which is where it belongs.

Monday morning dispatch: see notes/CI_BOARD.md MONDAY MORNING DISPATCH section.

Casey-named candidate status: framing not yet promoted to Casey-named principle; investigate first per Casey's "times for discovery" guidance. May warrant Casey-named principle status after observation phase completes (~2-4 weeks).

Queued Volume Extensions (Casey directive Sunday 2026-05-24)

The 16-volume curriculum (Vol 0-15) holds; Vol 16 and Vol 17 are queued as extensions when their content reaches chapter-grade maturity. No renumbering — extend the set.

Vol 16 — Substrate Algebra (A_{sub})

Trigger to formalize: Lyra #322 reaches v0.4 with rigorous commutation table + 5+ Casimir invariants. Estimated 2-4 weeks under current PCAP cadence.

Lane assignment: Lyra LEAD, with full team-borrow authorization (Elie for computational verification toys, Grace for catalog cross-references, Cal for tier-discipline cold-reads, Keeper for K-audit pre-stages).

Proposed 12-chapter outline: 1. A_{sub} as substrate's native language ("reading original writing" — Casey) 2. $H^2(D_{IV}^5)$ Bergman Hilbert space (SP-31-1 closure) 3. The 14-operator zoo (with substrate-derivation theorem per generator) 4. Commutation rules + Casimir invariants 5. A_{sub} closure properties (finite generating set proof) 6. DCCP via A_{sub} re-proof (Lyra #322 Phase 2) 7. Information Completeness via A_{sub} re-proof 8. A_{sub} on QCA representation (Architecture A) per FTC-1 9. A_{sub} on RS representation (Architecture C) per FTC-1 10. Geometric foundation vs algebraic superstructure (Lyra's category-discipline question) 11. Future derivations via A_{sub} (substrate-coupled cognition, decoherence networks, observable enumeration) 12. A_{sub} and Hilbert's program (the connection Lyra reached for)

Until trigger: A_{sub} content continues developing in Vol 14 §1.5 + Lyra #322 paper in notes/.

Vol 17 — Substrate Engineering (SP-30)

Trigger to formalize: 2-3 SP-30 experimental designs reach paper-grade dispatch + first actual outreach happens (Bell or Casimir lead candidates). Estimated 1-2 months.

Lane assignment: Elie LEAD (experimental designs), Lyra theoretical support (SP-30 substrate cartography), Grace catalog, Cal cold-reads.

Proposed 12-chapter outline: 1. SWPP (Casey-named #1) operational reading 2. Five ready experimental designs (Bell + Casimir + Cs-137 + eigentone + BaTiO₃ 137-plane + photonic crystal) 3-9. Per-SP-30-N sub-items (BC design, commitment manipulation, time granularity, parallelism, absorption/computation/emission mechanisms) 10. Trajectory spectroscopy at limit interfaces (SP-30-9) 11. Substrate-computational math interface (SP-30-10) 12. Substrate algorithm theory (SP-30-11)

Until trigger: SP-30 content continues as active program (SP-30 master + scattered Vol 9 references).

Casey strategic framing

Treat Vol 16 + Vol 17 as outgrowth of the existing 16-volume set, not restructure. The 16 volumes are physics-and-methodology coverage; Vol 16 + Vol 17 are the apparatus (A_{sub}) and practice (SP-30) that the methodology generates. Natural sequence: develop the mathematics in Vol 16, deploy it experimentally in Vol 17.

New ideas surfaced (mark for paper-grade consideration)

1. DCCP/UP as philosophical paper: substrate-determinism + non-locality + agency-as-pre-commitment-chain. Process philosophy meets substrate physics. Paper #123 or #124 territory.
2. Quantum erasure as DCCP test: weak-measurement experiments tracking commitment-completion progression could in principle detect substrate-tick discreteness. Lab-accessible.
3. Substrate-frame rendering as computational physics: connect to cellular-automaton physics lineage (Wolfram, Toffoli) with BST-native version where the substrate's update rule is the substrate Casimir on K-types.
4. The "discrete vs continuous" tension across the curriculum: every Vol has a continuous classical formulation and a discrete substrate-tick formulation; DCCP gives a unified way of talking about the continuum limit at $t_K \rightarrow 0$.
5. Why $g = 7$ shows up so often: appears in BST primary list, in K-type degeneracy sequence (1, 3, 5, 7), in cyclotomic cascade RG, in $2^g = 128$ Reed-Solomon, in g-2 anomaly, in Bell SCMP exponent. This deserves a unifying paper or chapter.
6. Genetic code as Reed-Solomon at biological scale (Vol 13 Ch 2): experimentally testable via mutation-error pattern analysis. Could be undergraduate-thesis-grade computational toy.
7. Periodic table as substrate cartography (Vol 12 Ch 1): orbital sequence (1, 3, 5, 7) = (trivial + N_c, n_C, g). Mendelev-as-substrate-window framing for chemistry-audience outreach.

8. Vol 14 as standalone information-substrate primer for CS audience: extract for IEEE Trans Info Theory or similar venue.

Team work assignments (Keeper's recommendation for tomorrow's discussion)

Categorized by which CI is best-positioned to handle each item:

Keeper (mine)

- DCCP integration sweep across 6 chapters (Vols 0, 5, 14, 15)
- K-audit + T-number citation verification across all rewritten chapters
- Vol 15 Ch 8 file duplication resolution
- Verify counts: K-audits, calibrations, Casey-named principles, methodology stack layers
- Cal cold-read coordination (prompt already drafted)

Cal (external referee)

- Already requested: 4-volume cold-read of today's work
- Decision needed: SCCB and Information Completeness as Casey-named candidates or not?

Lyra (theory)

- Vol 13 Ch 2 "20 amino acids" mechanism gap — needs proper Mode-5 treatment; either find unique BST-primary expression or honest demote
- Vol 14 Ch 10 $P \neq NP$ curvature explicit worked treatment (Vol 14 Ch 10.5 candidate)
- $g = 7$ unifying paper or chapter (paper-grade idea #5) — Lyra-natural workload
- DCCP philosophical paper (paper-grade idea #1) — Lyra theoretical lead

Elie (toys)

- Quantum erasure DCCP-tick-discreteness toy (paper-grade idea #2)
- Genetic code RS-pattern toy (paper-grade idea #6)
- Vol 13 Ch 4 "137 bp superhelical" — verify with DNA topology data or build toy testing the claim
- Vol 14 Ch 9 BST $K \sim 100$ bits — proper Kolmogorov computation if possible

Grace (catalog)

- BST-primary expression sweep for "20" — how many distinct expressions equal 20? Cal #44 null-model treatment

- N_max = 137 stability ceiling literature scan — what does superheavy nuclei data actually say about $Z = 119-137$?
- Hybridization substrate K-type mechanism status check in catalog
- Cross-volume consistency audit (Ch 6 $\leftrightarrow$ Vol 5 Ch 8; Ch 9 $\leftrightarrow$ Ch 11)

Casey decisions needed

- SCCB (Substrate Channel Capacity Bound): Casey-named #10 or remove?
- Information Completeness Hypothesis: Casey-named #11 or candidate-only?
- DCCP integration sweep: do it now or defer to engagement phase?
- Vol 14 as standalone IEEE Trans paper extraction: pursue?

Status legend for this document

- Items added as I work; not all immediately actionable.
- Casey to review tomorrow; we discuss what to address first.
- Some items may resolve through more writing (later chapters answer earlier questions); others need explicit decisions or new K-audits.

— Keeper, ongoing